

## Master Math for JEE Main & JEE Advanced

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### For JEE Main 2020 April

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## JEE Mains 2020 Jan Chapter wise Question Bank

## Calorimetry

Q1

When  $m$  gram of steam at  $100^\circ\text{C}$  is mixed with  $200$  gm of ice at  $0^\circ\text{C}$ . it results in water at  $40^\circ\text{C}$ . Find the value of  $m$  in gram .

(given : Latent heat of fusion ( $L_f$ ) =  $80$  cal/gm, Latent heat of vaporisation ( $L_v$ ) =  $540$  cal/gm., specific heat of water ( $C_w$ ) =  $1$  cal/gm/ $^\circ\text{C}$ )

जब  $100^\circ\text{C}$  पर,  $m$  ग्राम भाप को  $0^\circ\text{C}$  पर  $200$  gm बर्फ के साथ मिश्रित किया जाता है। यदि जल का परिणामी ताप  $40^\circ\text{C}$  हो, तो  $m$  का मान ग्राम में ज्ञात कीजिए।

(दिया है : गलन की गुप्त ऊष्मा ( $L_f$ ) =  $80$  cal/gm, वाष्पन की गुप्त ऊष्मा ( $L_v$ ) =  $540$  cal/gm., पानी की विशिष्ट ऊष्मा ( $C_w$ ) =  $1$  cal/gm/ $^\circ\text{C}$ )

7<sup>th</sup> Jan Evening

Sol

40

$$m_{\text{ice}} L_f + m_{\text{ice}} (40 - 0) C_w = m_{\text{steam}} L_v + m_{\text{steam}} (100 - 40) C_w$$

$$\Rightarrow 200[80 + 40(1)] = m[540 + 60(1)]$$

$$\Rightarrow 200 (120) = m(600)$$

$$m = 40 \text{ gm}$$

Q2

There are three containers  $C_1$ ,  $C_2$  and  $C_3$  filled with same material at different constant temperature.

When we mix then for different volume then we get some final temperature as shown in the below table.

So find value of final temperature  $\theta$  as shown in the table.

तीन पात्रों  $C_1$ ,  $C_2$  तथा  $C_3$  में समान पदार्थ अलग-अलग नितय तापमान पर भरा हुआ है। जब हम इनके अलग-अलग आयतनों को मिलाते हैं तो सारणी के अनुसार हमें मिश्रण के अन्तिम ताप प्राप्त होते हैं, तो सारणी में दिखाये अन्तिम ताप  $\theta$  का मान ज्ञात करो।

| $C_1$ | $C_2$ | $C_3$ | $t(^{\circ}\text{C})$ |
|-------|-------|-------|-----------------------|
| 1ℓ    | 2ℓ    | 0     | 60                    |
| 0     | 1ℓ    | 2ℓ    | 30                    |
| 2ℓ    | 0     | 1ℓ    | 60                    |
| 1ℓ    | 1ℓ    | 1ℓ    | $\theta$              |

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**50° C**

$$1\theta_1 + 2\theta_2 = (1 + 2) 60$$

$$\theta_1 + 2\theta_2 = 180 \quad \dots(1)$$

$$0 \times \theta_1 + 1 \times \theta_2 + 2 \times \theta_3 = (1 + 2) 30$$

$$\theta_2 + 2\theta_3 = 90 \quad \dots(2)$$

$$2 \times \theta_1 + 0 \times \theta_2 + 1 \times \theta_3 = (2 + 1) 60$$

$$2\theta_1 + \theta_3 = 180 \quad \dots(3)$$

and तथा  $\theta_1 + \theta_2 + \theta_3 = (1 + 1 + 1) \theta \quad \dots(4)$

from (1) + (2) + (3) से

$$3\theta_1 + 3\theta_2 + 3\theta_3 = 450 \Rightarrow \theta_1 + \theta_2 + \theta_3 = 150$$

from (4) equation  $150 = 3\theta \Rightarrow \theta = 50^\circ\text{C}$

समीकरण (4) से  $150 = 3\theta \Rightarrow \theta = 50^\circ\text{C}$



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